

GIS 2024 Model Exam

1.What are spatial relationships in a geodatabase?

- A. Relationships between tables in the database.
- B. Relationships between spatial features based on their geometry.
- C. Relationships between different versions of the geodatabase.
- D. Relationships between spatial data and non-spatial data.

Answer: B

2.Which of the following is not a common type of spatial relationship?

- A.Contains
- B.Overlaps
- C.Links
- D.Within

Answer: C

3.What is topology in a geodatabase?

- A. The study of spatial patterns in data
- B. The geographic extent of a dataset
- C. The spatial relationships and rules governing spatial features.
- D. The process of joining spatial datasets.

Answer: C

4.What is a spatial join in GIS?

- A. A process of combining spatial features based on their location.
- B. A process of joining tables in a geodatabase
- C. A process of visualizing spatial data on a map
- D. A process of exporting spatial data to a different format.

Answer: A

5.What are the two primary types of geodatabases in Esri's ArcGIS software?

- A. File geodatabases and personal geodatabases.
- B. Enterprise geodatabases and spatial geodatabases.
- C. Cloud geodatabases and local geodatabases.

D. Structured Query Language (SQL) geodatabases and NoSQL geodatabases.

Answer: A

6.Which type of geodatabase is suitable for single-user desktop applications?

A. Enterprise geodatabase

B. File geodatabase.

C. Personal geodatabase

D. Cloud geodatabase

Answer: B

7.What is a key characteristic of a personal geodatabase?

A. It supports versioning and replication.

B. It is stored as a single file on a local machine.

C. It requires a centralized server for data storage.

D. It is accessible over the internet.

Answer: B

8.Which type of geodatabase offers advanced capabilities such as versioning, replication, and multi-user editing?

A. Personal geodatabase.

B. File geodatabase.

C. Enterprise geodatabase.

D. Spatial geodatabase.

Answer: C

9.What is the primary purpose of tables in a geodatabase?

A. To store spatial data such as points, lines, and polygons

B. To provide a visual representation of spatial features on a map

C. To store attribute data associated with spatial features.

D. To perform spatial analysis operations

Answer: C

10.What is a spatial index in a geodatabase?

A. A table that stores information about spatial features

B. A data structure that enhances the retrieval of spatial data based on their location.

C. A reference system used to define the spatial properties of features.

D. A tool for visualizing spatial relationships between features.

Answer: B

11. What is a feature class in a geodatabase?

A. A collection of related spatial features that share the same geometry type.

B. A table that stores attribute data associated with spatial features.

C. A file format for storing raster data.

D. A data structure that organizes spatial features into layers.

Answer: A

12. What is the primary difference between a feature class and a feature dataset?

A. Feature classes store raster data, while feature datasets store vector data.

B. Feature datasets can contain multiple feature classes, while feature classes cannot.

C. Feature datasets are used for storing attribute data, while feature classes are used for storing spatial data.

D. Feature datasets are limited to one geometry type, while feature classes can contain multiple geometry types.

Answer: B

13. Aerial photographers use _____ to counteract the movement of the aircraft.

A. Special camera

B. Stereoscopy

C. Gyro-stabilizers

D. Digitizer

Answer: C

14. One is not types of aerial photograph, based on the direction /position of camera axis. Which is it?

A. Panchromatic

B. Vertical photographs

C. Oblique photographs

D. Focal length

Answer: A

15. What is the minimum location point/ points required to use of "triangulation" principle

A. Two

B. Three

C. Four

D. Six

Answer: A

16. _____ is parcel-based Land Information System.

A. Land information system

B. Land administration

C. Cadastre

D. Survey

Answer: C

17. A unit of land with homogenous tenure interests, having a unique owner/tenant, land class and use is referred to as a _____.

A. Cadastre

B. Land information system

C. Parcel

D. Land management

Answer: C

18. _____ is a public, up-to-date land information system (LIS) that efficiently supports public administration of real estate.

A. Parcel

B. Cadastre

C. Land information system

D. Surveying

Answer: B

19. _____ is the discipline of land surveying that relates to the laws of land ownership and the definition of property boundaries.

A. Cadastre

B. Cadastral surveying

C. Parcel

D. Land information

Answer: B

20. Which term refers to the amount of detail in a digital image?

A. File format

B. Pixel density

C. Metadata

D. Resolution

Answer: D

21. Which of the following is NOT a common application of multispectral satellite imagery?

A. Land cover classification

B. Vegetation health monitoring

C. Meteorological forecasting

D. Urban development analysis

Answer: C

22. What is the main advantage of hyperspectral imaging over multispectral imaging?

A. Higher spatial resolution

B. Captures a wider range of wavelengths

C. Lower cost

D. Limited data storage requirements

Answer: B

23. What is the primary advantage of digital image processing?

A. Higher cost compared to traditional methods

B. Faster processing speed

C. Lower quality of images produced

D. Limited applications in various fields

Answer: B

24. What is the primary goal of image enhancement?

A. To decrease the visual appeal of images

B. To complicate the interpretation of images

C. To make images more useful and visually appealing

D. To add noise to images

Answer: C

25. Which process corrects sensor errors and atmospheric effects in remote sensing images?

A. Geometric Correction

B. Radiometric Correction

C. Noise Reduction

D. Image Enhancement

Answer: B

26. What is the primary purpose of band rationing in remote sensing?

- A. To enhance the spatial resolution of the image
- B. To reduce noise in the image
- C. To highlight specific features by dividing the pixel values of one band by another
- D. To convert the image to grayscale

Answer: C

27. Which vegetation index is specifically designed to minimize the influence of soil background reflectance?

- A. NDVI
- B. EVI
- C. SAVI
- D. Green NDVI (GNDVI)

Answer: C

28. What is the primary goal of digital image analysis?

- A. To compress image files for storage
- B. To enhance the visual appearance of images
- C. To extract meaningful information from images
- D. To convert images from one format to another

Answer: C

29. Which of the following describes "morphological operations" in image analysis?

- A. Techniques for color correction
- B. Operations that process images based on shapes
- C. Methods for image compression
- D. Algorithms for image segmentation

Answer: B

30. In supervised classification, what is the role of training data, and how can the quality and quantity of this data affect the classification results?

- A. Training data is used to validate the accuracy of the classification results; higher quantity ensures overfitting.
- B. Training data helps in the initial clustering of data points; higher quality ensures faster processing.

- C. Training data is used to train the classification algorithm by providing known examples of different classes; higher quality and sufficient quantity lead to more accurate and reliable classification results.
- D. Training data is used for post-classification smoothing; higher quality ensures reduced processing time.

Answer: C

31. What is image segmentation in the context of remote sensing and GIS?

- A. A technique to classify pixels into different classes based on spectral characteristics
- B. The process of combining multiple images into a single, seamless image
- C. The division of an image into meaningful regions or objects
- D. The comparison of spectral characteristics over different time periods

Answer: C

32. Which image segmentation technique involves partitioning an image into segments based on the intensity of the pixels, where pixels with similar intensities are grouped together?

- A. Region growing
- B. Thresholding
- C. Edge detection
- D. Clustering

Answer: B

33. Which one is satellite-based positioning system?

- A. Levels
- B. Theodolite
- C. Total station
- D. Sputnik

Answer: D

34. One is not Global Satellite Navigation and Positioning System

- A. GLONASS
- B. Galileo
- C. GPS
- D. BeiDou

Answer: D

35. One is false about augmentation Systems (ASs)

- A. Ground-based AS provide a higher precision service than satellite-based AS

- B. Ground-based AS is designed to serve local area
- C. Satellite-based AS provide a higher precision service than ground-based AS
- D. Satellite-based AS is broadcasting to the users using geostationary satellites

Answer: C

36. How many Satellites constellation of GPS, GLONASS, Galileo, and Beidou respectively?

- A. 4, 3, 6, 8
- B. 6, 3, 3, 3
- C. 8, 6, 4, 4
- D. 3, 3, 3, 6

Answer: B

37. Which one is the simplest methods of measuring horizontal distance?

- A. Pacing method
- B. Taping method
- C. Stadia method
- D. EDM method

Answer: A

38. A measurement is made along a line that is inclined by a vertical angle of 30 deg as measured using a hand level & clinometer. The slope measurement is 756.52m. What is the corresponding horizontal distance?

- A. 378.26m
- B. 655.17m
- C. 436.78m
- D. 565.71m

Answer: B

39. One is hardly true about distance measurement by pacing method

- A. The length of a pace is varying with different persons
- B. The length of a pace is erratic with the speed of pacing
- C. The distance measured by pace will give more accurate result
- D. The length of a pace is fluctuating the slope of terrain

Answer: C

40. A line XYZ is measured on the slope in two segments. The first segment XY measures 824.45m and the second segment YZ measures 1244.38m. If the difference in elevation between points X & Y is

4.25m & that between Y & Z is 6.47m, determine the horizontal length of the measured line.

A. 2068.80m

B. 1492.72m

C. 1500.46m

D. 2186.08m

Answer: A

41. A stadia rod held at a distant point B is sighted by an instrument set up at A. The upper and lower stadia hair readings were observed as 1.300m and 0.900m, respectively. If the stadia interval factor(K) is 100, and the instrument constant(C) is zero, determine the horizontal distance of the line AB.

A. 30m

B. 40m

C. 10m

D. 20m

Answer: B

42. Suppose you measure a slope distance of 201.324m from A to B. you also measure the zenith angle at A of 122 degree 15'21" What is the corresponding vertical distance?

A. 107.45m

B. 319.01m

C. 170.25m

D. 391.10m

Answer: C

43. Except one all are true about rules for determine significant figures

A. All nonzero digits are significant

B. Zeroes between other significant figures are significant

C. Zeroes placed at the end of decimal numbers are significant

D. Leading zeroes to the left of the first nonzero digits are significant

Answer: D

44. Why the watershed management approach is appropriate means of natural resource management

A. Water is followed from upstream to downstream

B. Soil erosion is not followed from downstream to upstream

C. Degradation of upstream is cause for fragile downstream

D. To protect agroecology of the up and downstream

Answer: C

45. The main object of watershed management is

- A. To improve the living standard of the upstream community
- B. Using the watershed for irrigation purposes
- C. To develop more than one resource at a time
- D. To increase the influence of natural and manmade deathtraps

Answer: C

46. If you are assigned to perform irrigation suitability analysis for Anger watershed, which GIS function is appropriate to compute suitability analysis

- A. Neighborhood function
- B. Measurement function
- C. Network function
- D. Overlay function

Answer: D

47. From the Remote Sensing data resolution which one is used to identify the status of healthy and unhealthy crops.

- A. Spatial resolution
- B. Radiometric resolution
- C. spectral resolution
- D. Temporal resolution

Answer: C

48. From the following statement, one is correct about Precision farming.

- A. Decrease the duration of harvesting
- B. Regulate the amount of fertilizer and water applied at a given location
- C. Make the most of an energy waste
- D. Maximize nutrient-deficient sites.

Answer: B

49. Which multispectral image bands are suggested to assess crop conditions information

- A. Microwave band
- B. Near Infrared band
- C. Mid Infrared band
- D. Panchromatic band

Answer: B

50. The reason multi-criteria analysis is optimal ground water management is?

- A. Groundwater is dynamic resource
- B. Groundwater is found inaccessible region
- C. To evaluate and integrate different data layers
- D. To maximize groundwater discharge

Answer: C

51. To say one geographic area is suitable for agriculture or not, what criteria is needed to be fulfilled

- A. Availability and proximity of the river
- B. Availability of road and town
- C. Availability of fertile soil
- D. The intersection of stated criteria

Answer: D

52. Suppose that the Oromia Agricultural office went to study the drought status of the Borana zone, which satellite is mostly recommended to be used

- A. SPOT -XS
- B. MODIS -TERRA
- C. AVHRR
- D. ASTER -TERRA

Answer: B

53. Why the Geospatial data are the most important for agricultural and watershed management

- A. It gives real-time spatial information
- B. manage land more efficiently
- C. reducing impact and costs
- D. All

Answer: D

54. From the following one is used to develop a set of standards and specifications for how spatial data is accessed and distributed.

- A. HTTP
- B. GML
- C. ISO
- D. OGC

Answer: D

55. Of the following one is correct about the principle of open-source GIS

- A. License Must Be Specific to a Product
- B. The freedom to run the program, for any purpose
- C. Discrimination Against Fields of Study
- D. No Integrity of The Author's Source Code

Answer: B

56. In Distributed GIS, the function of client and server are integrated computing systems What does it indicate?

- A. The system is platform dependent
- B. The system is application dependent
- C. The client can only request service from the server and the server can only provide service
- D. Any computer could be both a server if it provides service and a client if it requests service

Answer: D

57. In Internet GIS Interoperability implies:

- A. Ubiquitous GIServices
- B. Flexible data approach between geodata and metadata scheme
- C. System cooperation process
- D. Reusability and compatibility of GIS software

Answer: C

58. WEB GIS is supporting the:

- A. Data compatibility system
- B. Application dependent function
- C. Unseparated set of GIS architecture system
- D. Visualized, and edited geodata in the browser.

Answer: D

59. In interactive web mapping at which phase symbols are attached to geospatial data

- A. Display
- B. Selection
- C. Display Element Generator
- D. Render

Answer: C

60. Of the following one is used for geospatial data processing on the server side as a service, and manages the incoming requests and replies with the proper map data back to the client side

- A. Interactive Web Maps
- B. Internet Map Servers
- C. ArcGIS online
- D. Web map

Answer: B

61. In Web GIS Architectures what is the role Common Gateway Interface (CGI)

- A. To display raster data
- B. limit communication between server and client
- C. Interpret communication between server and client
- D. Display vector data

Answer: C

62. From the following one is correct about advantage of Google earth engine

- A. Limited numbers of training samples
- B. developing new tools is easy
- C. Limited upload data
- D. Capable of big geospatial data analysis

Answer: D

63. From the following one is the most spatial database used for spatial data storage and to perform spatial Query

- A. XML
- B. GML
- C. Post GIS
- D. Open GIS

Answer: C

64. The process of designing, implementing, generating, and delivering maps on the World Wide Web is known as:

- A. Web client
- B. Web server
- C. Web mapping

D. Web browser

Answer: C

65. Which one is the main benefit of using RS and GIS in forestry?

A. Reducing the cost of timber extraction

B. Enhancing public awareness about forests

C. Providing accurate, up-to-date, and comprehensive data for effective forest conservation and management

D. Increasing recreational use of forests

Answer: C

66. How does airborne LiDAR scanning facilitate forest stock estimation?

A. By providing recreational maps for tourists

B. By collecting detailed forest structure data rapidly over large areas

C. By increasing the market price of timber products

D. By reducing the cost of forest management

Answer: B

67. Why is monitoring forest biomass using remote sensing important?

A. To enhance tourism activities in forests

B. To understand carbon storage and assess forest productivity

C. To increase the recreational use of forests

D. To reduce the cost of forest management

Answer: B

68. What role does LiDAR data play in habitat mapping and biodiversity conservation?

A. Enhancing recreational facilities

B. Providing detailed canopy height models and forest structure data

C. Increasing the market value of forest products

D. Promoting tourism in protected areas

Answer: B

69. What is the primary purpose of thematic mapping in forestry?

A. Enhancing recreational activities

B. Understanding and managing forest ecosystems

C. Increasing timber production

D. Promoting tourism

Answer: B

70. What does the variable 'Forest Cover' measure?

A. The economic value of forest products

B. The extent of land covered by trees

C. The number of tourists visiting the forest

D. The number of recreational facilities in the forest

Answer: B

71. What are the primary functions of FMIS in forest management?

A. Enhancing Forest tourism

B. Spatial mapping and inventorying of forest resources

C. Increasing recreational facilities

D. Promoting urban development

Answer: B

72. One of the following data definition language statements is used to generate new table in the relational database management system. Which one?

A. Create

B. Alter

C. Drop

D. None

Answer: A

73. Which of the following is not set of data manipulation language?

A. Where

B. And

C. Or

D. Create

Answer: D

74. Peoples have the right to live in fresh environment (free of pollution). Therefore, Waste disposal location must be far away from household by some distance as recommended by expert and large organization such as FAO. Which spatial analysis function is used to zoning waste disposal location by the given distance?

A. Clip

- B. Buffer
- C. Extract
- D. Spatial Overlay

Answer: B

75. Sudan is country which located at neighbor of Ethiopia and they do have common boundary with one another. In Topology such relationship is called as _____

- A. Adjacency
- B. Connectivity
- C. Containment
- D. Network

Answer: A

76. The techniques of combining two spatial layers and producing a third one from them is called _____

- A. Clipping
- B. Buffering
- C. Extracting
- D. Spatial Overlaying

Answer: D

77. Among element of Database Manipulation Language statement, which one of the following statement is wrong in extracting data from database?

- A. WHERE - Extracts only those records that fulfill a specified criterion.
- B. AND - Displays a record if both the first condition and the second condition are true.
- C. OR - Displays a record if either the first condition or second condition or second condition is true.
- D. AND - Displays a record if the first condition or the second condition is true.

Answer: D

78. Which of the following GIS Architecture/Work Flow is well sequenced?

- A. Capture data, Store data, Query data, Analyze data, Display data, Present data
- B. Capture data, Query data, Store data, Analyze data, Display data, Present data
- C. Query data, Capture data, Store data, Analyze data, Display data, Present data
- D. Capture data, Query data, Store data, Analyze data, Present data, Display data

Answer: A

79. One is not macro-economics factors in terms of urban sprawl causes

- A. Economic growth
- B. Globalization
- C. Population growth
- D. European integration

Answer: C

80. Which one is odd about the capabilities of GIS in urban planning?

- A. design of buildings
- B. visualization
- C. spatial analysis
- D. spatial modeling

Answer: A

81. _____ refers to the usefulness or enjoyment a consumer can get from a service or good.

- A. Urban sprawl
- B. Urban planning
- C. Utility
- D. Urban growth

Answer: C

82. _____ is a perspective that serves to guide the structuring of decision rules in MCE procedure.

- A. Decision
- B. Objectives
- C. Criterion
- D. AHP

Answer: B

83. _____ essentially determine the quality of the information derived from remotely sensed data.

- A. Image interpretation
- B. Accuracy assessment
- C. Supervised classification
- D. Unsupervised classification

Answer: B

84. _____ can handle any type of data distribution and trend, as well as account for any type of anisotropy in the interpolation method.

- A. IDW
- B. Kriging
- C. Spline
- D. Universal kriging

Answer: B

85. Which spatial resolution is most important for studying urban areas?

- A. Low spatial resolution
- B. Medium spatial resolution
- C. High spatial resolution
- D. None of the above

Answer: C

86. From the following which one is true about local function?

- A. A function that work on all cells within the raster
- B. Each cell is processed only based on its own values
- C. The output value of each cell is a function of the entire grid
- D. A function which works on cell locations within zones

Answer: B

87. _____ is the procedure by which criteria are selected and combined to arrive at a particular evaluation, and by which evaluations are compared and acted upon.

- A. Criteria
- B. Decision rule
- C. Pair wise comparison
- D. Multicriteria evaluation

Answer: B

88. Which of the following is an example of an error in the source of GIS data?

- A. Scanning errors
- B. Digitizing errors
- C. Errors in maps used by surveyors
- D. Reprojection errors

Answer: C

89. Which spatial data error is created due to the lack of consideration for the gap between two polygons during digitization?

- A. Dangling node
- B. Sliver polygon
- C. Pseudo node
- D. Weird polygon

Answer: B

90. Among the listed geoprocessing tools, which one is employed to eliminate boundaries or nodes between adjacent polygons or lines sharing the same attribute value?

- A. Merge
- B. Intersect
- C. Dissolve
- D. Union

Answer: C

91. If a node extends beyond the entity it's meant to connect to during digitization, what type of spatial data error occurs?

- A. Closed polygon
- B. Overshoot
- C. Undershoot
- D. unclosed polygon

Answer: B

92. Which of the following best describes the primary purpose of the Hydrology tools in GIS analysis?

- A. Modeling the flow of electricity across a surface.
- B. Analyzing the distribution of vegetation across a surface.
- C. Analyzing the movement of wildlife across a surface.
- D. Modeling the flow of water across a surface.

Answer: D

93. If you intend to evaluate the suitability of land for irrigation in the given district, with soil composition being a significant factor, which GIS geoprocessing tool should you utilize to extract the soil data specifically for your area of interest?

- A. Clipping
- B. Overlay
- C. Merge

D. Intersect

Answer: A

94.What is the primary purpose of using buffers in GIS analysis?

- A. To analyze patterns in temporal data by creating data clocks.
- B. To determine the proximity of features within one or more feature classes.
- C. To store geographic data containing dates and times as a tracking layer.
- D. To create an area around a feature at a uniform distance for analysis purposes.

Answer: D

95.What is the first step in a typical GIS analysis workflow?

- A. Implementing the analysis
- B. Choosing an appropriate data
- C. Examine the results
- D. Framing the problem as a question

Answer: D

96.Which of the following is a recommended solution for storing extremely large datasets that need to be accessed by many people in a GIS project?

- A. Storing data on a local drive
- B. Distributing data across multiple USB drives
- C. Storing data on a central server
- D. Using external hard drives for data storage

Answer: C

97.What feature properties can be extracted using the Triangulated Surface toolset?

- A. Temperature, precipitation, and humidity
- B. Aspect, slope, and contour isolines
- C. Population density, urban areas, and rural settlements
- D. Vegetation types, soil composition, and land cover

Answer: B

98. Which component of spatial data represents the actual spatial location or shape of geographic features?

- A. Geometric Coordinates
- B. Attribute Data
- C. Spatial Relationships

D. Non-Spatial Coordinate

Answer: A

99. What type of spatial relationship is concerned with features being next to each other?

A. Adjacency

B. Connectivity

C. Containment

D. Proximity

Answer: A

100. If a given ellipsoidal have an equatorial and polar radius of 6,378.206 and 6,356.584 m respectively. Which one of the following is NOT correct?

A. The curvature of its polar radius is 6,399.9

B. Flattening 0.00339

C. First eccentricity 0.0822

D. First eccentricity 0.003407546

Answer: D

101. One of the following is NOT among the positioning satellites designed to derive 3D geodetic datums:

A. Lunar Lather Ranging (LLR)

B. Satellite Lather Ranging (SLR)

C. The GNSS constellation

D. Landsat Satellites

Answer: D

102. Which one of the following is NOT required so as to derive the World Geodetic System?

A. Intercontinental links

B. Measuring gravity data

C. Defining suitable ellipsoid

D. Deriving land use/cover data

Answer: D

103. A vertical distance from a location on the Earth's Surface and the geoid is

A. Orthometric Height

B. Normal Height

C. Geodetic Height

D. Dynamic Orthometric Height

Answer: A

104. One of the following is NOT used to specify Earth ellipsoid:

A. Semimajor axis

B. Flattening

C. Total mass of the ellipsoid

D. Digital Images

Answer: D

105. Which one of the following is an Earth-centered unified geodetic reference system?

A. Adindan Universal Transverse Mercator Zone 37 N

B. World Geodetic System 1984

C. North American Datum 27

D. Clark 1886

Answer: B

106. Which one of the following is WRONG about Geodetic Datums?

A. Are references to which surveying & mapping results are referred

B. Provide initial data for all kinds of surveying and mapping work

C. Define the radius, flattening, semi-major and semi-minor axis of the earth

D. Derived not necessarily based on a network of precisely surveyed control points

Answer: D

107. Which one of the following is a modern method of computing coordinates of horizontal control points?

A. Traversing method

B. Triangulation method

C. Trilateration method

D. Satellite GPS based method

Answer: D

108. A contractor sent out six surveyors to measure a distance between two points of a road marked on the ground. Each of the six surveyors came up with the following six different values: 265.25m, 265.15m, 264.90m, 266.04m, 265.50m and 261.22m. Assuming these values are equally reliable and that variations result from accidental errors, what is the Most Probable Value of the distance measure?

A. 1588.06m

B. 264.6767

C. -0.36

D. -0.05143

Answer: B

109. Which one of the following is correct about the referenced ellipsoid of the Earth?

A. Bulged at the poles and flattened at the equator

B. Bulged at the equator and flattened at the poles

C. Is geoidal in shape

D. Spherical in shape

Answer: B

110. One of the following is made to establish horizontal and vertical positions of arbitrary points:

A. Topographic survey

B. Control survey

C. Photogrammetric survey

D. Cadastral surveys

Answer: B

111. Assume a Bench Mark at point A has an elevation of 2000m, the back sight value is 10.2m. What is the elevation of point B at a foresight value of 1.2ms?

A. 2009m

B. 1091m

C. 9m

D. 2019m

Answer: A

112. The Azimuth of point B from A is 1120. What would be its bearing?

A. S680E

B. N680W

C. S2480W

D. N2480E

Answer: A